

Project Report

Telecom Customer Churn Analysis

Introduction:

Customer churn is a major challenge in the telecom industry, directly affecting revenue and customer lifetime value. This project analyzes customer data using Python and SQL to identify key drivers of churn, quantify its impact, and recommend strategies to improve retention.

The analysis reveals that contract type, tenure, pricing, and service usage significantly influence customer churn. High-risk customer segments were identified and actionable recommendations were proposed.

Objective:

- Measure overall churn rate
- Identify factors driving customer churn
- Segment customers based on risk
- Estimate revenue loss
- Provide data-driven business recommendations

Dataset Description:

The dataset used in this project was sourced from Kaggle and contains customer-level data from a telecom company.

It includes detailed information across the following categories:

Category	Variables
Demographics	Gender, SeniorCitizen, Partner, Dependents
Account Information	Tenure, Contract
Services	InternetService, OnlineSecurity, TechSupport, Streaming
Billing	MonthlyCharges, TotalCharges, PaymentMethod
Target Variable	Churn (Yes/No)

The dataset provides a comprehensive view of customer behavior, service usage, and billing patterns, making it suitable for churn analysis and business decision-making.

Data Source:

- Dataset Name: Telco Customer Churn Dataset
- Platform: Kaggle
- Link: <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

This dataset is widely used for customer churn analysis and represents realistic telecom business scenarios.

Data Preparation:

The dataset required preprocessing before analysis:

- Converted TotalCharges from string to numeric format
- Removed missing and invalid values
- Encoded Churn as binary (1 = Yes, 0 = No)
- Created a new feature:

$$\text{Average Charges} = \text{TotalCharges} / \text{Tenure}$$

- Handled division errors (e.g., tenure = 0)

Key Performance Indicators (KPIs):

Metric	Value
Total Customers	7043
Churned Customers	1869
Churn Rate	26.54%
Average Monthly Charges	64.76
Average Tenure	32.37 months

SQL Analysis & Findings:

1. Overall Churn Rate:

- The overall churn rate is **26.54%**
- Indicates a significant customer retention issue

2. Contract Type Analysis:

- Customers with month-to-month contracts have the highest churn rate
- Customers with long-term contracts (1–2 years) show lower churn

Insight: Short-term commitment increases the likelihood of churn.

3. Tenure-Based Segmentation:

- Customers with **tenure < 12 months** show the highest churn
- Long-tenure customers are more stable

Insight: Customer retention is weakest during the early lifecycle stage.

4. High-Risk Customer Identification:

Customers with:

- Month-to-month contracts
- Tenure < 12 months
- Monthly charges > 70

show the highest churn probability.

Insight: This segment should be prioritized for retention strategies.

5. Payment Method Analysis:

- Customers using **electronic check** show higher churn
- Customers using **automatic payment methods** show lower churn

Insight: Payment convenience plays a role in customer retention.

6. Service Impact Analysis

- Customers without Tech Support have higher churn rates.

Insight: Value-added services contribute to customer loyalty.

7. Revenue Loss Analysis

- Estimated monthly revenue loss due to churn: **139130**

Insight: Churn has a direct financial impact on the business.

8. High-Value Customer Churn:

- Some high-paying customers are also churning

Insight: Losing premium customers can significantly affect revenue.

9. Multi-Factor Analysis:

- Churn varies across combinations of: Contract type and Internet service

Insight: Customer churn is influenced by multiple interacting factors.

Python-Based Analysis:

Python was used for exploratory data analysis and visualization:

- Identified churn distribution across key variables
- Analyzed patterns in:
 - Contract type
 - Tenure
 - Monthly charges
 - Service usage
- Generated visual insights using Seaborn and Matplotlib

Key Insights:

- Month-to-month customers are most likely to churn
- New customers (less than 1 year) have the highest churn risk
- Higher monthly charges are associated with increased churn
- Lack of support services increases churn probability
- Payment methods influence customer behavior

Business Recommendations:

- **Promote Long-Term Contract:** Offer incentives and discounts for annual plans
- **Improve Customer Onboarding:** Focus on the first 6–12 months of customer experience
- **Target High-Risk Customers:** Use segmentation logic to identify and retain at-risk users
- **Bundle Value-Added Services:** Encourage adoption of Tech Support and Security services
- **Optimize Payment Methods:** Promote automatic payment systems to reduce churn

Tools & Technologies

- **Python:** Pandas, NumPy, Seaborn, Matplotlib

- **SQL:** MySQL
- **Techniques:** Data Cleaning, EDA, KPI Analysis, Customer Segmentation

Conclusion:

This project successfully analyzed customer churn using Python and SQL, identifying key drivers and high-risk segments. The findings provide actionable insights that can help telecom companies reduce churn, improve retention, and minimize revenue loss.